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THE CRAFT OF SCIENTIFIC WRITING

Third Edition



Actually Sitting Down to Write

The writing aspect of scientific research is exhausting... I have rewritten many parts of papers four to six times, restructuring the entire organization, before I finally became satisfied.

—Hermann Helmholtz

No one can tell you the most productive way to get your words onto paper. Perhaps you can learn something from imitating the habits of professional writers, but the actual act of sitting down to write is individual. What has worked for Joan Didion or James Agee might not work for you.

Scientific writing is hard work. Granted, it's not as physically exhausting as swinging a pick or as mentally demanding as solving a nonlinear differential equation, but it requires concentration and patience. Moreover, the solutions are not exact. You don't draft a document and sit back and say, "Perfect." No matter how many times you revise a document, there will always be phrases that won't sound right, as well as sentences where you feel compelled to state about ten details at once.

Scientific writing is often lonely work. Although you

might brainstorm ideas in a group or solicit sections of a document from other people, the process of sitting down to write calls for periods of solitary confinement. Just because scientific writing is difficult does not mean it is drudgery—not at all. Writing a scientific document demands energy. You have to convey complex ideas and images. Strong scientific writing also demands imagination. You have to detach yourself from your work and place yourself in the position of your audience. Moreover, you will find that writing a strong document challenges not only your writing skills, but also your scientific and engineering skills. When you write to inform or to persuade, rather than to impress, you see your own theories and experiments as your audience sees your theories and experiments. You become a critic of your own work. As Francis Bacon said, “Reading maketh a full man, conference a ready man, and writing an exact man.”

So where do you get started? First, you need something to write about, either an idea for a proposal or results from your work. Although you don’t have to wait until every result has been recorded, mapped, and plotted before you begin writing, you should not start writing a scientific or engineering document that has no destination.

The second thing that you need is an understanding of your constraints, particularly who your audience is and what the format will be. First, you must have an idea about who will read the document: what they know about the work, why they will read the document, and how they will read the document. Without an audience, you won’t know what background information to include, which words to define, or how detailed to make your illustrations. You also should have an idea about the format. For instance, how long will the document be—two pages? twenty pages? two hundred pages?

Let’s assume that you’ve got something to write about and that you understand your constraints. Let’s say

that you've spent the last fourteen months testing solid particles as a new heat transfer medium for solar power plants. The simplicity of the idea intrigues you. You envision a solar central receiver plant in a desert with hundreds of mirrors surrounding a central receiver. Falling through the receiver is a stream of silica sand that collects solar energy reflected from the mirrors and stores that energy in a tank below.

You've not only got visions; you've got results from simulation experiments in which concentrated sunlight has heated silica sand particles to more than 1000°C [Hruby, 1984]. You've also finished a computer model that shows the concept's thermal performance to be about 75 percent, a performance competitive with current central receiver designs [LaJeunesse, 1985].

The constraints of format and audience you also know. The audience is challenging because it is so varied: Department of Energy (DOE) officials, solar engineers, and utility engineers. Within this mix of audience, there are political considerations for the document. Over the past few years, funding for the solar energy program has dropped because current designs of solar central receivers have not proven efficient enough for utilities to invest in them. Management in the solar program is looking for some new ideas, and you see solid particles as a good one. Unfortunately, many officials at DOE want a quick-fix solution—they're antagonistic to something long-term such as solid particles. Nonetheless, you see solid particles as the best chance that solar energy has for competing with fossil fuels.

Your supervisor sees something too because she is eager for you to publish your work. "Have you written your report?" she asks you.

"No, not yet," you say.

Your supervisor stares at you for a moment. She has a casual look about her that's deceptive. Her office suddenly feels hot. "I'll be seeing something soon," she says.

"Soon," you say. "Soon."

You go back to your office where Hank Wilson is waiting. Hank Wilson wants to talk about the Braves. You remind yourself that during the last players' strike, you vowed never to watch another baseball game again. You remind yourself that your old heroes—Aaron, Cepeda, and Niekro—no longer play the game, and that a lot of the people who do play the game receive more money in a year than you'll earn in your lifetime.

Hank starts in about last night's game where the Braves came up with two runs in the bottom of the eighth to come back against the Reds. "It started with a couple of singles, both grounders with eyes, and then McGriff ripped a double to the gap."

Hank's story reels you in for a moment, but you remember your vows. The only problem is that the Braves have a three-game lead in their division, and as long as you can remember, you have been a diehard Braves fan. As a kid, the reason was silly: You wanted them to win the series so that you could go to a ticker-tape parade. As you grew older, you wanted the Braves to win a series for the simple reason that you had been waiting so long. Through the years of Aaron and Niekro and Murphy, you waited, reading boxscore after boxscore, staying up nights to catch radio broadcasts from the west coast, relishing each win, dying a little with each loss. You hesitate a moment, watch the light flicker in Hank's eyes, but don't give in.

Even if you wanted to talk about baseball, you can't. You've got results, you've got recommendations, and you've got a report to write. What should you do? You can't write here with Hank talking about baseball. The library, you think. "Catch me on it later, Hank," you say. "I've got to write my report."

You are almost out of the building when the department secretary stops you. "Al put something in your mail slot," she says.

"I don't want to look," you say.

"Then don't," she says.

For one instant, you think about moving on, but curiosity gets the better of you. Al is the department manager. In your mail slot is the program's *Quarterly Report*. It's your turn to review it. The *Quarterly Report* is the pride and joy of some paper pusher in the DOE. It contains almost nothing technical, only management things: milestone charts, procurement summaries, fiscal plots, and contract distributions.

"By tomorrow," she says.

"Tomorrow?"

"Don't look at me. That's what Al said."

"But I've got to write my report."

"That's fine," she says. "As long as this gets reviewed by tomorrow." The department secretary smiles at you. She has a wicked smile.

You go back to your office. Hank Wilson stares bewildered as you walk by. You try to ignore him. At your desk, you begin reading. You are only halfway through the standard DOE foreword when you catch yourself staring at the words and sentences, but not comprehending them. You sit up in your chair, go back to the first sentence, and begin again. Thirty minutes into your reading, a utility engineer from Arizona calls. His company is studying solar energy options, and he wants to know the thermal efficiencies of solid particle receivers. You promise to send him something tomorrow.

You push through the *Quarterly's* milestone charts and fiscal sheets and begin checking the procurement summaries when an engineer from analytical modeling stops by. She wants to adapt your computer code to solve a problem in coal combustion. You force a smile and invite her in. She has several questions about your code—questions you try to answer off the top of your head, but can't. So you dig through your old notes and printouts.

By the time she leaves, it is five-thirty. You are tired and hungry. What now? You decide to go home and return here tonight, when no one is around and you can take the phone off the hook and write.

You get home and there's a bill from the power company. It's a bill you're sure you've already paid. You throw it in the wastebasket. Then you pick it out. There's something frightening about throwing away an unpaid bill. You decide to forget the bill and eat out. You need a good meal, you think. After all, you've got a lot of writing in front of you. You go to Pedro's for chili rellenos. While you eat, you read today's newspaper. You glance at the sports section. Glancing at the boxscores is okay, you decide. Reading about baseball is not actually watching baseball, and that was your vow: never to watch another baseball game. The Braves play in Houston this weekend. Houston has never been kind to the Braves. You know the reason, of course. It's the Astrodome. Baseball is supposed to be played outdoors.

It is seven o'clock by the time you finish eating and get back home. You're a little tired and you decide to rest a bit—just lie down on the couch and think about what you're going to write.

When you wake, it's quiet and dark outside. You look at the clock: 11:43. The day is lost.

Getting in the Mood

Scientific writing is a craft that requires preparation. It is difficult to jump right into writing a document. Most professional writers have disciplined schedules. Although you as a scientist or engineer cannot alter your entire lifestyle just to get one document written, you can make some simple adjustments.

The first thing you should do before writing is to

clear your mind. You have to put aside your personal problems and the bills you have due. You have to think about the work and the audience. One way to clear your mind is through sleep. Hemingway used this method. His writing day began each morning after eight hours of sleep. Another good way to clear your mind is through motion: walking, running, mowing the lawn. Wright Morris walks twice a day, before and after writing. Harper Lee plays golf. Max Apple and John Irving are avid runners. Mowing is okay, but you've got only so much lawn. Driving lets you daydream, but driving and daydreaming at the same time is dangerous. Running, walking, and bicycling are the best. They give you the chance to think about the structure of the document, play a strategy through in your mind, and see whether it makes sense. I prefer a steady run, about thirty to forty minutes, not so long that I'm exhausted, but long enough that I begin daydreaming. After taking a run or walk, I don't eat much more than a snack. Then I shower and slip into something casual.

The second thing you need is a block of free time. For drafting a document, a good writing block is one to four hours. Even after sitting down at your desk, you usually need ten minutes or so just to get in the groove of actually writing sentences. Don't think, though, that you can continue writing for the whole day. After a while, your concentration fades, and your writing efficiency diminishes.

Finding a block of time at work is often difficult. There are so many interruptions: phone calls, e-mail messages, visitors. To make the writing process efficient for yourself, you should eliminate those interruptions. You should shut the door, unplug the phone, mute the electronic mail messages, and concentrate on getting your thoughts onto paper. You will find that absolute silence in your room is not necessary. White noise is okay. So is soft instrumental music. However, each interruption, no matter how short, will cost you an additional five or ten

minutes of writing time because it takes you that much time to get back into the document—to find your place again and begin writing sentences that add to what you’ve already done.

Some distractions are created, rather than imposed. A few years ago, computer games, especially the game of bridge, posed a problem for me. If that game was on the computer, I felt a constant force tugging me away from my writing. Just a window away was a freshly dealt hand. After struggling for a few months to achieve a balance between work and play, I went cold turkey one day and deposited all my computer games in the trash. The result was that my writing productivity increased dramatically.

Finally, you have to prepare yourself mentally for the task ahead. Scientific writing is hard work. It is not something that you can casually begin after a heavy meal or a couple of drinks. Many scientists and engineers mistakenly think of professional writers as free spirits who write perhaps two days a week and play the other five. In this imagined schedule, writers spend most of their time fishing, traveling, or going to bull fights. Actually, most professional writers are diligent workers. Hemingway, for instance, followed a schedule of writing for eight hours a day, six days a week.

Instead of worrying about the way that professional writers spend their leisure time, you should consider the way that professional writers spend their writing time. For instance, you should look at how professional writers structure their sentences and paragraphs. Don’t, however, choose writers from the eighteenth century whose language is outdated, or writers such as Faulkner or Joyce whose styles are luxuriant. Rather, choose writers whose styles are crisp and straightforward. Richard Rhodes, who authored *The Making of the Atomic Bomb*, is an excellent choice. So is Annie Dillard, author of *Teaching a Stone to Talk*, or David Simon, author of *Homicide*.

Writing First Drafts

The next day, you quickly take care of the small things at work: your critique of the *Quarterly Report*, a memo to the utility engineer in Arizona. You leave work early and eat a light meal at home. You wash the dishes, put in a load of laundry, and go out for a slow run—something to clear your mind. You run out beyond the subdivision, out into cattle land where you're surrounded by barbed wire and brown grasses. There is still daylight, and you are thinking about what you are going to write.

After you finish your run, you shower and drive back to the office. No one is there. It is almost seven o'clock. Somehow the place seems different. Maybe it's the stillness, or just the quiet. The only sound is the hum of the drinking fountain. You sit at your desk with your spiral notebook of results in front of you. You are in the mood.

You turn on the computer and stare at the blank screen. You finger the keyboard, but do not type anything. You are still thinking. The steady, expectant hum of the computer bothers you so much that you turn toward your other table and grab a pencil. You need something written down before you can start on the computer. You write a title for your paper:

Solid Particle Solar Receiver Design

Tiresome, you think. Too many big words in a row. You erase and try again.

Solid Particle Receiver Design

That's too confusing. Most people wouldn't know that you are talking about solar energy. More important, the research isn't really the design of a solid particle receiver, but whether solid particles could work as the heat transfer medium in a solar central receiver. You write another title:

Using Solid Particles in Solar Central Receiver Systems

It's not great, but it will get you started. You put down your pencil and pick up your pen. Ink looks so much more professional than lead. You rewrite the title in ink and then write your name below the title. The rest of the page is blank. It stares back at you. Suddenly, you feel tired. You lean back in your chair and rub your eyes. You loathe first drafts. You would do almost anything instead of writing a first draft: fill out your taxes, clean the bathtub, watch a stock car race on television.

Come on, you tell yourself. You've wasted enough time. You must start. You must write something down. An outline, you think. That's what you were always taught. So you write one:

- A. Introduction
- B. Experiment
- C. Computations
- D. Conclusions

What about a summary. You write "Summary" above "Introduction." What letter should you assign it? You already used A for "Introduction." You decide to make it Z. You're not happy with Z, but you see little choice besides crossing out the other four. You wish that you had stayed with the pencil. Now you look at your outline.

- Z. Summary
- A. Introduction
- B. Experiment
- C. Computations
- D. Conclusions

Not much help you think.

You decide to try something else. You write "Summary" at the top of a blank page, then "Introduction" at the top of another page, and then "Experiment," "Computations," and "Conclusions" at the top of three more. You pick up your five outline pages as well as your title page, bunch them into a stack, and then place them back down. Now you're getting somewhere.

You put aside the title page and start with the summary page, but you remember that an informative summary is the last thing that you write. So you put it aside and look at your introduction. You start writing a sentence about how the solar central receiver program began in 1976, but then you stop. What's an introduction supposed to do? Introductions are supposed to prepare readers for what's ahead. How can you prepare someone for something ahead if you don't know what that something is. You put the "Introduction" page aside as well.

So far, things aren't going well.

For some reason, you are restless. It is eight o'clock and you are hungry again. You go down the hall and get a candy bar. You know that candy bars are bad for you, but you're working hard tonight and you deserve a treat. After you finish the candy bar, you feel tired. You consider calling Hank Wilson to casually ask whether he watered the office plants today, although you know he hasn't. You have the hope that he might volunteer the Braves score. Then you think about going home and crawling into bed and sleeping until you can't sleep anymore.

You have lost the mood.

Don't deceive yourself. The first draft of a document is difficult. In the first draft, you juggle your topic and your writing constraints with all three elements of style. What should you do to get through a first draft? The first thing you should do is take a few deep breaths and prepare yourself for the road ahead. If the document is long, you're not going to finish the draft in one sitting. If you do, chances are that the draft won't be any good. In writing, you have to be realistic. It has taken you several months to do your work. It's also going to take some time to finish the document—days in some cases, weeks in others.

Now for the actual act of writing the first draft. Should you use a pencil, pen, or computer? As John Gar-

dener [1983] said, that question is much deeper than it sounds. The real question, he says, is how do you get words onto paper. Regarding the surface question of which writing instrument to use, you should use whatever it takes. A computer certainly has the most advantages. On a computer, it's easy to add text, delete text, bring in text from other documents, and incorporate illustrations. Still, some people feel more comfortable beginning on sheets of paper, at least for the brainstorming stages. Writing on sheets of paper has the advantage of allowing you to see the document's entire strategy on your desk without having to scroll through the document on the computer. Also, many people feel uncomfortable sitting before a computer and just thinking, which working on the initial stages of a document requires you to do.

What about the deeper question of how you get words onto paper? The answer is the same: whatever it takes, short of plagiarism. Should you use an outline? Every English teacher you've ever had has probably told you to use one. That fact alone should make you suspicious. The answer to the question is a qualified "yes." A strong outline is helpful for long and complicated documents. A strong outline gives you a structure before you begin writing. That way, in your early drafts, you have to juggle only two elements of style—language and illustration—rather than three. To be effective, though, your outline must be strong. What makes for a strong outline? Detail. A strong outline not only lists your section headings, but also your subsection headings, notes about each section, and some sentences or phrases you might use. Outlines by professional writers usually don't look like the neat little ABCD things your English teachers put on the blackboard. Professional outlines, such as the one shown below, are long and spread out and above all, filled with the kinds of things that make it easy for you to write a first draft.

OUTLINE
 USING SOLID PARTICLE RECEIVERS
 IN SOLAR CENTRAL RECEIVER DESIGNS

Audience

The principal audience is DOE headquarters. They have a general knowledge about solar central receivers, but not about solid particle research. Some DOE members are antagonistic to the solid particle idea (they want a quicker solution).

The secondary audience includes solar engineers at the utilities and national laboratories. These people are knowledgeable about solar central receivers and mainly want to know whether the solid particle idea can work.

Introduction

Identity of research:

This report will discuss using solid particles as the heat transfer medium in a solar central receiver. Requirements for such a heat transfer medium include

- (1) Heat transfer medium must withstand temperatures up to 800°C
- (2) Convective losses from medium must not be too high (define "too high")
- (3) What else?

Reasons that research is important:

- (1) Diversifies solar energy applications by obtaining higher temperatures than those obtainable from current designs
- (2) Allows for high-temperature experiments and modeling

Background material (heat transfer media):

- (1) *Water/steam*. Advantages: utilities familiar with; test data available. Disadvantages: not efficient enough; difficult to control in two-phase flow.
- (2) *Molten Salt*. Advantages: relatively high heat capacity; high storage capacity; single-phase fluid. Disadvantages: freezing in pipes; corrosive nature.
- (3) *Liquid Sodium*. Advantages: high heat transfer capacity. Disadvantages: low storage capacity; safety problem.
- (4) *Solid Particles*. Advantages: very high heat transfer capacity (define "very high"); absorbs radiation directly; transfers heat directly. Disadvantages: abrasive nature; utilities less familiar with.

Mapping of Discussion

Discussion

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Note that this outline has parts of sentences interspersed among the details. If you are writing an outline and suddenly come up with an idea for a sentence or transitional phrase, it makes sense to include that sentence or phrase in your outline. If not, you risk the chance of forgetting the sentence or phrase when you begin the first draft. What if someone else asks to see the outline, though? Won't all of these details and questions and sentence parts confuse them? Probably. Sometimes supervisors ask to see the outline of a document to check the progress of that document. In such cases, you should clean up your outline so that your supervisor can see the strategy. As it turns out, your "cleaned up" outline will often look like the ABCD outline that your English teacher taught you. Still, your "working" outline should be in a form that helps you write the first draft. If your working outline includes green ink for illustration ideas and red ink for ongoing questions, so be it.

How do you actually move from the outline to the first draft? Let's first assume that you have a strong outline and that you are in the mood to write. Now, there are two basic approaches to writing the first draft: the rabbit's and the turtle's.

Rabbits hate first drafts. They despise juggling the constraints of writing with all the elements of style. So, in a first draft, they sprint. They write down everything and anything. Rabbits strap themselves in front of their computers and finish their drafts as quickly as possible. Unfortunately, their first drafts are horrendous, sometimes not much better than their outlines. Still, they've got something. They've put their ideas into a document, and they're in a position to revise.

Turtles, on the other hand, are patient. Turtles accept the job before them and proceed methodically. A turtle won't write down a sentence unless it's perfect. In the first sitting, a turtle begins with one sentence and

slowly builds on that sentence with another, and then another. In the second sitting, a turtle then goes back to the beginning and revises everything from the first sitting before adding on. It usually takes a turtle several sittings to finish a first draft, but the beginning and middle are smooth because they've been reworked so many times. Revision then entails looking at the document from an overall perspective and smoothing the ending.

Which type of writer should you be? The answer is whatever type works for you. Actually, few professional writers are strictly rabbits or turtles. Most professional writers are a combination. If you become stuck in the first draft and your writing slows to a stop, you should become a rabbit so that you can finish. On the other hand, if you're working on a section that establishes the organization for the rest of the paper, you should be a turtle; otherwise, you might have to start over.

Which section of a paper should you begin writing? The introduction or the discussion? This question is difficult. On the one hand, you psychologically need some kind of introduction before you can begin the discussion. On the other hand, the discussion is the easiest section to write first. My feeling is that you should be a rabbit on the first draft of your introduction and then write the discussion and conclusion at your normal pace. After finishing the conclusion, you should go back and rewrite the introduction. What about your summary? A descriptive summary you can write first, but an informative summary you want to write last. One good way to draft the informative summary is to wait until you have a polished draft of the text. At that point, you can highlight the most important sentences and illustrations of the draft. With a little transition and background added, these highlighted sentences and illustrations can serve as a first draft for the informative summary.

The key to writing first drafts is momentum. Because

scientific writing is so textured (in other words, because each sentence depends so much on the ones around it), you need to sustain momentum during your first draft. Otherwise, you'll lose your train of thought. How do you sustain momentum? Well, here are a few habits that you can borrow from professional writers:

First, set realistic goals. How large a goal depends on how much of a rabbit or turtle you are. If you're a rabbit, then a realistic goal for a single sitting might be five pages. If you're a turtle, a realistic goal might be one page. Realistic goals are important. Psychologically, you want to end each sitting with a feeling of accomplishment.

Second, end your sittings by writing into the next section. That way you'll find it easier to start writing the next time you sit down. You will have a little momentum rather than a blank page. The fiction writer André Dubus claims that when he ends a sitting, it is mid-sentence, mid-thought.

Third, watch what you eat. Writing makes you restless, which makes you hungry. One way to counteract that is to sip on a tall glass of water while you write. If you must eat, don't eat foods that will make you sleepy. Also avoid salty foods that make you thirsty or foods that require two hands such as bananas (you have to peel them). You want something that you can reach with one hand. Celery and carrot sticks are probably the best, but you can munch on them for only so long. Raisins aren't too bad; neither are unsalted sunflower seeds. Since the advent of breadmakers, I've been opting for slices of fresh bread.

Finally, when you finish a draft, store it on your computer under a separate file name. Having a clean version that you can print out at any time gives you a psychological advantage when you start revising the original. You won't worry about being caught empty-handed in case your supervisor needs a clean draft right away.

How do you write a first draft when you're pressed for time? There are really no shortcuts in writing a first

draft outside of abandoning your friends and family and locking yourself in a room until it's done. Actually, when you're pressed for time, the one part of the writing process that you don't want to cut short is the first draft. If anything, you should spend extra time on your first draft because you can't afford to choose a structure that is inappropriate. Choosing an inappropriate structure could cause you to have the document in pieces on your desk when the deadline strikes.

What do you do if you've got writer's block? Writer's block is one of those bits of folklore that many scientists and engineers talk about, especially when they're having problems writing. "The words just aren't flowing," they'll say. "I must have writer's block." Well, the words rarely "flow" for anyone, professional writers included. Most professional writers struggle the same way you do. They struggle with paragraphs and sentences and words. So what is the source of this term "writer's block"? When writing a document, there are certain things, besides distractions, that cause you to stop.

First, you can't think of the right word. Not having the right word happens to everyone. Many times it's on the tip of your tongue. In such a case, a possible solution is to pull out a dictionary and spend two or three minutes looking. If you don't find the word after three minutes, write down the name of your favorite baseball player (mine is Roberto Clemente) and keep going. Your subconscious will think of the right word later.

Second, you can't find the sentences to express an idea. In this situation, chances are that you haven't quite grasped that idea yet. A possible solution is to skip two lines and keep writing. Let the idea simmer. Maybe you can think about it on your next run or walk or exercise activity. Besides getting you in the mood to write, exercise is an excellent activity for helping you sharpen your ideas.

Third, you hear voices. When many scientists and

engineers sit down to write, they hear voices, critical voices: their eighth grade English teacher, their department manager, the theorist from Oxford who complains bitterly about everyone else's writing but doesn't write so well himself. These voices inhibit you—you just can't make your pen write another word. A possible solution is to put on your favorite classical music and turn up the volume until it drowns the voices. Bach's *Brandenburg Concertos* and Mozart's piano sonatas work well for me.

In scientific writing, there really is no such thing as writer's block. Once you have done the work, the ideas are there. Writing those ideas is more mechanical than inspirational. For scientists and engineers, the real block occurs when the ideas won't come, but that block is a block in science and engineering, not in writing.

Revising, Revising, Revising

Revision is the key to strong scientific writing. Revision is the difference between a title such as

Using Solid Particle Receivers
in Solar Central Receiver Systems

and a title such as

Using Solid Particles as the Heat Transfer Medium
in Solar Central Receivers.

For the work described in this chapter, the second title is more precise. You weren't deciding whether solid particle receivers could work in a solar central receiver system. Rather, you were deciding whether solid particles could work in a solar central receiver. This difference is important, but this difference might not have been apparent when you were staring at the blank computer screen of a first draft.

In your first draft, you have to juggle too many ele-

ments of style to give proper attention to any one. Revision gives you the luxury of considering on any single draft only the parallel structure of your headings or the conciseness of your language or the clarity of your illustrations. Scientists and engineers, perhaps more than other professionals, understand the importance of revision. In their experiments, scientists and engineers depend on trial and error. In their computations, scientists and engineers use many iterative techniques to arrive at solutions. Just as those techniques are important to science and engineering, so is revision important to scientific writing.

Many scientists and engineers hold the misconception that great writers don't have to revise. These scientists and engineers assume that great writers think only great thoughts and then effortlessly write them down in a prose that needs no reworking. Well, maybe there have been some great writers such as D. H. Lawrence whose first drafts could go straight to press, but Lawrence and writers like him are in the small minority. For every D. H. Lawrence, you can find ten writers who were constant revisers: Leo Tolstoy, Ernest Hemingway, Anton Chekhov, Ralph Ellison, S. J. Perelman, Carolyn Chute, E. B. White, Flannery O'Connor, Raymond Carver, Evan Connell. Raymond Carver claimed that he never took fewer than ten drafts to finish a story. If great writers such as Raymond Carver required ten drafts to smooth their writing, just think how many revisions the rest of us need. In my own writing, I average about five pages a day. Unfortunately, they're all the same page. It's unusual for me to have a sentence that was written on the first draft make it to the final draft unchanged. For most of us, the key to successful writing lies in working hard on our revisions, not in conjuring magic on our first drafts.

What does revision entail? Before you begin revising a document, you should obtain some distance from it. After finishing a first draft, spend a day working on

another project. Better still, hike in the mountains or have a night on the town. You need a few hours, sometimes a few days, away from your first draft so that you can effectively revise. Things have to settle. During this time, ideas for changes will probably come to you. Write these ideas down, but don't go back to your draft.

Let's assume that you have some distance between you and your draft. What next? An important thing for you to do is to change your personality. You should not have the same personality when you revise as you did when you wrote the first draft. Whether you worked as a turtle or a rabbit on the first draft, you were erecting the document. In the revision stage, you have to chisel and polish what's on the page.

To become a successful reviser, you have to become a good reader. How do you become a good reader? One way is to listen to other good readers. Just as no two people write the same way, no two people read the same way. Some people are particularly attuned to language. Others are sensitive to structure. Listening to good readers review a document will help you strengthen your own reading. A second way to become a good reader is to practice. Work hard on your critiques of other people's documents. Be specific. Don't just tell these people that something is unclear. Show them why you became confused. Also, don't just mark weaknesses in a document; mark the aspects that you find successful. You will find that telling someone what is strong in a document is as difficult as pointing out the things that are weak. How quickly you mature as a writer will depend on how quickly you mature as a reader.

What's the best method of revising? There is no definitive answer. Still, there are some techniques that many successful writers adopt:

One thing to do is to change the look of your document when you revise. If you drafted your document on a com-

puter, try revising on a hard copy. If you drafted your document in your office, try revising the document in the library or conference room. In other words, make the revision process different from the drafting process.

Second, try to work through large chunks in each sitting. Revising large sections of your document in single sittings makes your document smooth. You'll see gaps in logic and faults in structure. During your first few revisions, you'll be writing a lot—filling in spots. As with the first draft, set realistic goals for yourself on each sitting. In later revisions, you should have fewer and fewer corrections, and therefore you can read larger chunks. Ideally, you'd like to reach the point where you can read through the entire document in a single sitting or a single day.

Third, get some distance between each revision. Avoid making multiple passes through a large document on the same day. If you read a document or section over and over, it's easy to go stir-crazy and start changing things for no reason. How many times should you revise a document? For your documents to really sing, you should revise until you are nitpicking over language: a word here, a comma there. In your later drafts, try reading the most important sections aloud. You'll be surprised at how many things you'll catch with your ear. Although you want to get some distance between drafts, you don't want to get too much distance. Allowing a draft to sit on a desk for months certainly gives you distance from the draft, but also saps the momentum from the project. You'll feel compelled to update the science and engineering in the document, which will make the revision seem more like a first draft.

Finally, solicit criticism of your writing. Soliciting criticism is an excellent way to revise. No matter how you try to separate yourself from your work, you will take certain things for granted. For instance, in your report about solid particles, maybe you forgot to mention the composition of your solid particles in the summary (your solid

particles were silica sand). That detail would be an easy oversight for someone engrossed in the work, but one that a fresh reader would quickly catch. If you are going to solicit criticism, you should be willing to make changes. This caution might sound silly, but you wouldn't believe how many times someone has asked me to review a document and then been appalled when I actually suggested revisions.

Once you solicit criticism of your writing, you should be prepared to accept that criticism. Don't be defensive when people whom you respect speak harshly about your writing. They're not attacking you. Actually, when people become harsh over your writing, it's a compliment of sorts. It means they're frustrated—they're genuinely interested in your work and they want your writing to inform more efficiently. If your critics are indifferent, that's the occasion to be upset.

How do you know when to incorporate criticism and when to dismiss it? This question is difficult. You shouldn't give a rubber stamp to every suggested change, even if the suggestions come from a good critiquer. Rather, you should weigh all criticisms. If you think a criticism is valid, incorporate it. If you think a criticism is "off the wall," at least mull it over. Maybe there was a deeper problem within that section or paragraph that the critic couldn't articulate.

Don't seek too many critics. You'll get confused. On a five-year plan for one of its solar programs, the Department of Energy decided that everyone in the program should review the document. In effect, the Department of Energy wanted all the readers to become the writers. The result was chaos. Because the report was general in its original scope, the two hundred critiques only served to push it into two hundred different directions. For several years, the Department of Energy haggled over draft after draft of that report, wasting thousands of man-hours, before finally scrapping the whole document.

This example raises the question of how you should approach collaboration on a document. Collaboration has both advantages and disadvantages. The principal advantage is that working in a group broadens the range of ideas that the document can incorporate. Collaborative writing also allows the group to draw from the various editing strengths of the members, such as one person's ability to see the big picture or another's ability to smooth transitions between ideas or yet another's ability as a proofreader. The principal disadvantage of collaborative writing is an inherent lack of continuity. With multiple writers contributing to the text, you can have inconsistencies in language, particularly in word choices, average sentence lengths, and degrees of sentence variety. In an ideal collaboration, you optimize the advantages and minimize the disadvantages.

One way to accomplish that goal is to solicit everyone's ideas and strategies at the outlining stage, but then have one person be the lead writer for the project to insure that the document maintains continuity. In the revision stage, the group would again have input. For this collaborative scheme to work, the group should respect the writing ability of the lead writer, and the lead writer should respect the editing comments of the group. This scheme is not the only way to achieve a successful collaboration. More important perhaps than the collaboration scheme is the interaction of personalities. Each person in the group should realize that the final product will not read exactly as he or she would desire. In other words, everyone in the group has to bend a little. That flexibility does not mean that group members should tolerate illogical strategies or discontinuous language, but that group members should accept that there are multiple ways to organize a section and achieve sentence variety.

What happens to revision if you're on a tight schedule? In many projects, the time allotted for the writing is

just too short. You might have begun with good intentions (for instance, scheduling a month to write the report), but then you lost twelve days in February because of a late shipment, five days in April because of a sick technician, three days in May because of a vacuum leak, and another three days in July because the computers were down. The month that you scheduled for writing has slipped to one week. So what do you do? Well, as stated in the previous section, you should spend enough time on your first draft to secure a structure that works. Then you've got to revise efficiently.

Revising a document is much the same as leveling a piece of land. In your first couple of revisions, you wield a pick for groundbreaking as you edit on the section and subsection level. Then in your remaining revisions, you use a rake for smoothing, as you progress from rearranging paragraphs to polishing sentences to nit-picking over words. When time is short, you can't afford to revamp the structure in your last couple of revisions. If you do, you might improve the overall grade a little, but you'll unsettle the soil a lot. Also, when pressed for time, you should reduce the time between revisions. In this reduction, you should find some absorbing activity such as running or walking or perhaps a light meal away from the office to separate yourself from the previous draft.

At some point in the process, you have to say "enough"—enough to outside suggestions, enough to revisions. You have to decide that your document is successful and then you have to finish it.

Many scientists and engineers think that perfection can be achieved in writing. Except perhaps for the ten commandments, there is no perfection in writing (and even the ten commandments suffered revision between Exodus and Deuteronomy). Words are not exact substitutes for thoughts. There will inevitably be something that you will want to change: some sentence that won't sound

right on Wednesday, another sentence that won't sound right on Thursday.

Although there is no perfection in writing, there is success. For your sanity, you have to find a point at which you stop drafting. Evan Connell once said that he knew he was finished with a short story when he found himself going through it and taking out commas, and then going through it again and putting commas back in the same places. The documentation of strong science and engineering deserves that amount of care. Although you can't achieve perfection in your writing, you should still strive for it. Strive for perfection, but be content with success. One of the beauties of writing is that you don't stop learning. With each document, you improve your craft.

Finishing

It is two o'clock in the morning. Your report has gone through the laboratory's sign-off and you have finished proofing it for what seems like the hundredth time. You are exhausted, but you correct the last two typos and print out the final copy. No one is in the office. You wish someone was—anyone. They wouldn't have to read your report, just look at it, admire how clean and neat it is. You'd tell them how smooth the writing was, maybe read aloud the title and summary.

Finishing a large document is taxing work. By your final draft, the words seem dead on the page. You don't really read them. You only see yourself reading them. Although the final proof of a manuscript is often tedious, you can't let up. A small mistake such as a misspelled word will unsettle your readers and undercut the authority of your work. Finishing a paper is much the same as finishing a baseball game. Some teams, when they're ahead, let up during the last few innings. They play sloppily, sometimes so sloppily that they lose their lead. Some

writers are the same way. They work hard on the first few drafts, and then let up on the final drafts, allowing typos to pull down their work.

Should you rely on a computer spell-checker to find those typos? "Rely" is the wrong word here. Spell-checkers are wonderful for catching some misspellings such as "recieve" instead of "receive." Some spell-checkers even notice if you repeat a word such as "in the the chamber." However, spell-checkers miss a number of misspellings, such as when the misspelling is another word ("to" rather than "too"). Spell-checkers also do not catch errors such as missing words ("in chamber" rather than "in the chamber"). For these kinds of errors, you need a person rather than a computer. In other words, you should use a spell-checker, but that spell-checker should supplement, rather than replace, your own careful proofreading.

Besides being hard work, the completion of a large document is often thankless work. By the time your document appears in print and people comment on it, the feelings that you have for the writing have usually been lost. You are probably absorbed in your next project. That's the way the writing game is. Don't expect satisfaction in your writing to come from other people. You'll be sorely disappointed. Satisfaction in your writing has to come from within. It is the realization that you've done good work.

You make a final copy of your report and place it in the mail for the print shop. You are euphoric and exhausted at the same time. It is over. You leave the lab. The night is cool, and the moon glimmers behind some thin clouds. You think you hear opera. You get in your car—a maroon and white Monte Carlo with a bad muffler—and drive past the laboratories, past the open fields tinted purple and blue by the night. At a stop sign, your muffler wakes a neighborhood dog. You wish that you could wake everyone and tell them what you've done.

You're tired, but you can't go to bed—not just yet. You need some kind of celebration. Pedro's, you think. You'll order a chili relleno and read the sports pages. The Braves are headed for the playoffs, and this time you've got a good feeling that they're going to win.

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